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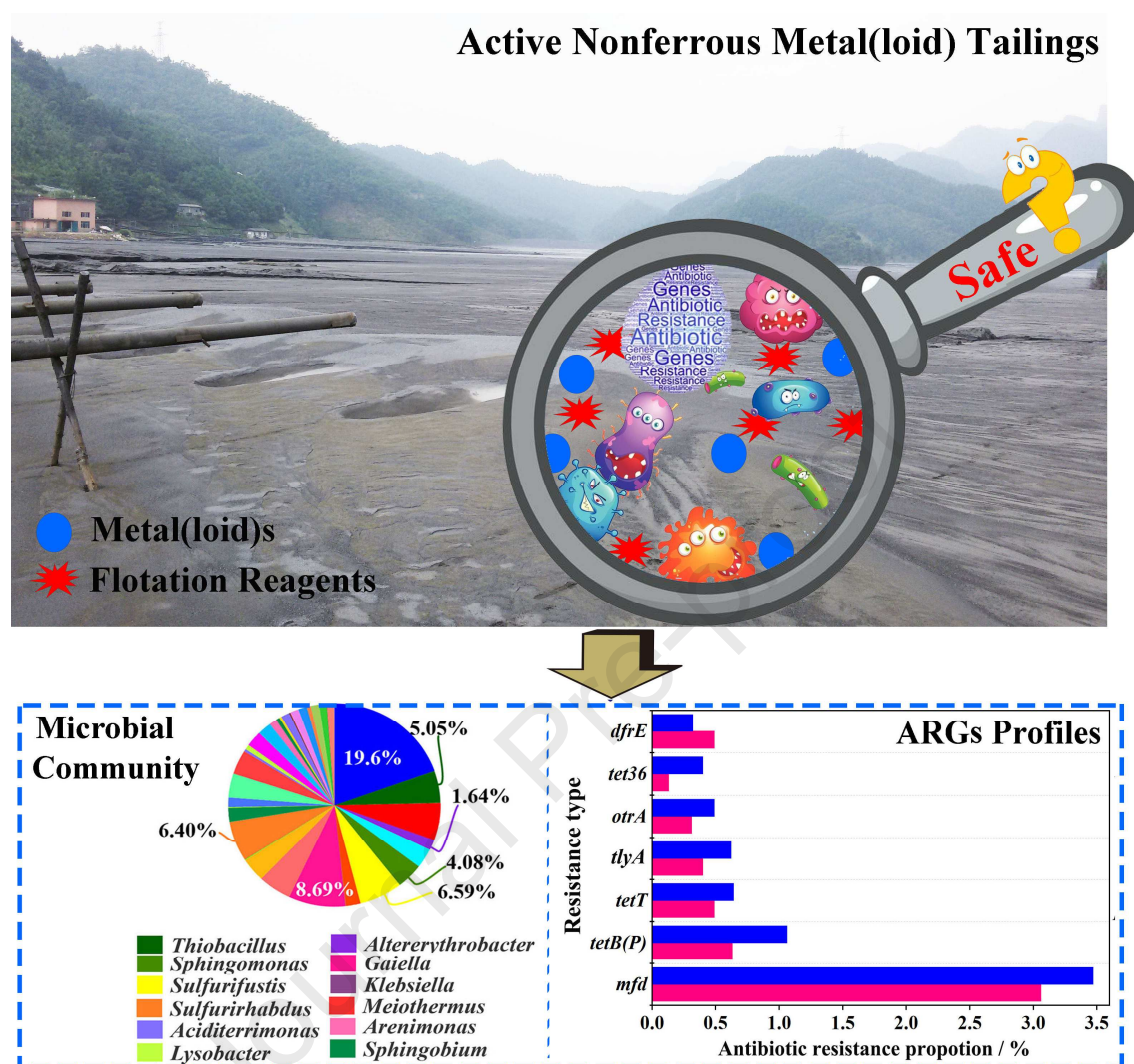
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Graphical Abstracts



Metagenomic exploration of multi-resistance genes linked to microbial attributes in active nonferrous metal(loid) tailings

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ABSTRACT

Mine tailings sites are considered as a continuous source of discharged metal(loid)s and residual organic flotation reagents. They are extremely toxic environments representing unique ecological niches for microbial communities. Mine tailings as a source of multi-resistance genes have been poorly investigated. Metagenomic analysis for four active nonferrous metal(loid) tailings sites with different environmental parameters was conducted. The abundance of *Thiobacillus*, able to tolerate acidity and showing iron- and sulfur/sulfide oxidation capacities, was significantly different ($p < 0.05$) between acid and neutral tailings sites. Correlation analyses showed that Zn, Pb, TP, Cd, and Cu were the main drivers influencing the bacterial compositions. Multi-metal resistance genes (MRGs) and antibiotic resistance genes (ARGs), such as *baca* and *copA*, were found to be co-selected by high concentrations of metal(loid)s tailings. The main contributors to different distributions of MRGs were *Thiobacillus* and *Nocardioides* genus, while genera with low abundance ($< 0.1\%$) were the main contributors for ARGs. Functional metabolic pathways related to Fe-S metabolism, polycyclic aromatic hydrocarbons (PAHs) degradation and acid stress were largely from *Altererythrobacter*, *Lysobacter*, and *Thiobacillus*, respectively. Such information provides new insights on active tailings with highly toxic contaminants. Short-term metal(loid) exposure of microorganism in active nonferrous metal(loid) tailings contribute to the co-occurrence of ARGs and MRGs, and aggravation of tailings acidification. Our results recommend that the management of microorganisms involved in acid tolerance and metal/antibiotic resistance is of key importance for in-suit treatment of the continuous discharge of tailings with multiple metal(loid) contaminants into impoundments.

Keywords: Typical active tailings; Metagenomic; Antibiotic resistance genes; Co-selection of metal(loid)s

1 Introduction

The active tailings (i.e. receiving continuous mine waste deposits) contain generally ~30 % solids (Blowes et al., 2003; COM, 2011). They are characterized by mixtures containing metal(loid)s and flotation reagents (FRs), including also mineral compounds formed by metal(loid)s-FRs complexes (Luan and Yao, 2002). In China, the mining activity produced 42% of total industrial metal(loid)s in 2013, and flotation reagents quantities that corresponded to three times more than those of pesticides in agricultural fields (Liu et al., 2018). Thus, active tailings have been classified as major sources of hazardous materials, threatening dramatically natural ecosystem functions (Bruneel et al., 2008; Hudson-Edwards, 2016). Microbial communities play a critical role in mobilizing and immobilizing metal(loid)s in active tailings (Bruneel et al., 2008; Ludovic et al., 2010). Microorganisms have the capacity to transform metal(loid)s into more migratory forms and degrade FRs to generate leachates containing bioavailable metal(loid)s (Cravo-Laureau et al., 2017; Liu et al., 2018), which are able to migrate contamination in the tailings surroundings, including farmlands and inhabited areas (Chao et al., 2016; Zhu et al., 2018).

Due to the massive usage of antibiotics, antibiotic resistance genes (ARGs) are wide spread within microorganisms and therefore ARGs are considered as emerging environmental threat to public health (Zhao et al., 2019; Zhao et al., 2020). It has been demonstrated that microorganisms harboring ARGs are selected in mine tailings (Chen et al., 2019; Yan et al., 2020). Because ARGs are associated with metal resistance genes (MRGs; Chen et al., 2013), it has been suggested that the ARGs dissemination is influenced by metal(loid)s content (Chen et al., 2019; Sun et al., 2020). Indeed, MRGs and ARGs have been often found associated and carried on the same mobile genetic element (plasmid, integron or transposon) (Bagade et al., 2016). Therefore, tailings are considered as ARGs sources that pose a global threat to public

health and food security because ARGs hamper the success of antibiotic treatments over a range of bacterial pathogens (Chen et al., 2019). However, patterns of MRGs and ARGs in metal(loid)s contaminated tailings are still poorly studied.

Guangxi (China), the most predominant nonferrous mining area in the world, is in karst geological landform facilitating the metal(loid)s migration (Rademaekers et al., 2011). The Guangxi tailings have high concentration of metal(loid)s and FRs, posing threats to surroundings, where are more serious than many other areas in the world (Liu et al., 2018; Zhu et al., 2018). Thus, these active tailings impose a challenge for environmental management, including the mitigation of pollutants (metal(loid)s and FRs), and microbial resistance (such as MRGs and ARGs), which may spread to humans *via* tailings dust and wastewater discharge (Chen et al., 2019).

A field investigation was conducted across four typical active nonferrous mine tailings at Guangxi (two acid and two neutral) to identify the factors responsible for the variability in microbial diversity and the presence of ARGs and MRGs. Several factors, including bacterial composition, environmental factors, and FRs, were analyzed to determine how the tailing conditions affect the resistant genes and functional metabolism. We hypothesized that active tailings may impose a selection pressure on microbial communities. Also, with short-term exposure to contaminants may result in the emergence of MRGs and ARGs along with genes involved in microbial metabolism which are specially adapted to extreme conditions of the tailings. To address these hypotheses, we collected the surface samples (top 10 cm) of the four active tailings sites. The diversity of MRGs and ARGs, and the microbial metabolism were investigated by metagenomic analysis. Additionally, environmental factors (pH, C/N/P/S, concentration of metal(loid)s and FRs) were determined. Multivariable analysis was performed in order to determine the key drivers influencing microbial diversity and metabolism, as well as

the presence of MRGs and ARGs.

2 Materials and methods

2.1 Field survey, sampling and physical-chemical analyses

Tailings samples were collected from four active impoundments sites in Guangxi, China (Fig. 1). The tailings sands stockpiled at these four tailings sites were mainly derived from Sn, Pb-Zn, and Pb-Zn-Sb mining and smelting processes (Table S1). Samples were collected in June 2016, and all the sites were surveyed following an EU standardized international sampling protocol (Grøn et al., 2007). Five to twenty plots per site were divided according to the locations of the tailings discharge pipelines and surface area. The coordinates and elevation of each site were recorded in situ with the GPS instrument (ZL166, ZL electronic technology, China) and were standardized to the WGS84 ellipsoid for visualization and analysis using Google Earth free software.

< insert Fig. 1 >

A nest sampling design was implemented at each plot, and 2×2 m quadrats were randomly placed. At each site, a composite sample consisting of 10 - 40 surface (0- to 10-cm depth) sub-samples was collected from each quadrant, bulked, and homogenized in the field. The numbers of sub-samples for each site was dependent on each site's surface area. In total of twelve samples (three parallel samples for each tailings) were subjected to 16S RNA sequencing, and one composite sample from each tailings was subjected to metagenome sequencing. Five hundred grams of composite sample at each site was collected for DNA extraction and physical-chemical analysis. The composite samples were transported in cooled boxes filled with ice within 2 days, and then stored at -20°C at the laboratory of China University of Geosciences (Beijing). Samples for physical-chemical analysis were air-dried

and sieved at 100-mesh size (0.149 mm, US standard). pH, total organic carbon (TOC), total nitrogen (TN), total phosphorus (TP), and total sulfur (TS) were defined followed by the China National Agricultural Technology Extension Center (2006). Concentration of metal(loid)s was determined after digestion with HNO₃-HCl-HF (5:3:2, v/v/v, GR) in a microwave unit. More detailed information were showed in Supplementary Information files. Physical-chemical parameters are summarized in Table S1.

2.2 Shotgun metagenome sequencing and data processing

Metagenomic DNA were extracted in triplicate from each composite sample, and then pooled into a composed DNA sample for sequencing. The EDTA buffer of the DNA Isolation Kit (UltraClean, MOBIO, USA) was used to dilute and eliminate potential contamination from metal(loid)s during DNA extraction. To determine the diversity and composition of bacterial communities in the active metal(loid) tailings, sequencing analysis was performed at the Illumina shotgun metagenome sequencing platform as described previously (Liu et al., 2019). The universal primer set 338F/806R amplified the V3-V4 region of the bacterial 16S rRNA gene, and an 8 bp barcodes was used for sample identification (Liu et al., 2016). The sequencing data were denoised with the FLASH and Trimmomatic software to obtain valid reads from raw data, and total of 34870 quality reads were remained after trimming for each sample (Table S2). An operational taxonomic units (OTUs) cluster analysis was conducted using Usearch software (version 7.0 <http://drive5.com/uparse/>), and then annotated based on the SILVA 16S rRNA bacteria database using a 70% confidence threshold (Chen et al., 2013; Zhou et al., 2016). The technical replicates for sequencing analysis were conducted for the active tailing's samples. The MicroPITA (microbiomes: Picking Interesting Taxonomic Abundance) analysis were used to select the target tailings sites with the most diverse and

dissimilar based on the Bray-Curtis dissimilarity matrix and Shannon a-diversity index.

Two strategies (16S rRNA gene sequencing and metagenomic analyses) were employed to reveal the microbial composition and metabolic functions in T1 and T4 tailings sites, representing the extreme acid (T1) and neutral (T4) active tailings, respectively. Shotgun libraries were conducted on the same composed DNA from the corresponding tailings samples and paired-end shotgun metagenome sequenced. To obtain a high quality data-set, the raw sequencing data were filtered to remove the low quality, deduplication, and adapter contamination reads using Seqprep and Sickle software. Subsequently, the high-quality reads were co-assembled using Multiple_Megahit method (option assembly; minimum length 300 bp) (Li et al., 2015). The gene-coding sequences (minimum length 100 bp) were predicted on the assembled contigs using MetaGene software, generating a total of 16.4 M gene-coding sequences. Sequences that could be not annotated were discarded, and subsequent analysis was performed on the metagenomes that were evenly sampled at the minimum annotated reads per sample. Sequences were annotated by comparing against the BLASTP (BLAST Version 2.2.28+) database (e-value threshold = 10^{-5}), and classified into taxonomic groups based on the NCBI non-redundant (nr) database (Altschul, 1997). The random shotgun sequencing generated 12 GB high quality dataset for each metagenome. All the genes with NCBI-nr hits were compared to the Kyoto Encyclopedia of Genes and Genomes (KEGG) databases. Metabolic pathways were then constructed for these NCBI-nr hits with the NCBI-nr, EggNOG, KEGG, and comprehensive antibiotic resistance databases. The detailed method for the PCR amplification of MiSeq sequencing analysis and calculation of relative abundance are provided in the Supplementary Information.

2.3 Statistical analyses

All significant statistical analyses were generated using IBM SPSS statistics 21.0 and RStudio free software 1.1.453. Differences of environmental parameters, MRGs and ARGs were statistically analyzed using one-way ANOVA and student's *t* tests, and $p < 0.05$ was considered significant. To explore the relative contribution of taxa to microbial metabolism and antibiotic resistance, nonparametric Kruskal-Wallis analysis of variance ($p < 0.05$), similarity percentage analysis, and multiple comparisons with Bonferroni correction were used, which has been described by Ofek-Lalzar et al. (2013). Correlations between the microbial relative abundance and environmental factors were analyzed by Spearman correlation analysis. Relative abundance data for MRGs, ARGs, functional metabolism pathways and OTUs were used as the input data for contribution analysis. The co-occurrence patterns between ARGs (normalized abundance) and MRGs (relative abundance of KOs related to metals) were explored using network analysis by Gephi software 0.9.2 with a 0.5 correlation coefficient.

3 Results and Discussion

3.1 General characteristics of tailing bacterial communities

The physical-chemical characteristics of the studied active tailings were determined at the surface layers (Table S1). The active tailings sites presented similar physical-chemical parameters, except for pH that was significantly different ($p = 0.01$; Table S1). This result combined with the fact that ore dressing and smelting in the tailings sites follow different processes (Luan and Yao, 2002; Kashani and Rashchi, 2008), suggested that the acidification phenomenon might be due to the use of distinct flotation reagents having different physical-chemical properties. The biodiversity of bacterial communities was determined by MiSeq sequencing. After sub-sampling each sample to an equal sequencing depth and

clustering, 748 OTUs were obtained with the number of OTUs ranging from 245 to 428 per sample (Table S2). The Good's coverage for the observed OTUs ($99.88\% \pm 0.03\%$, mean $\pm$ s.d.; Table S2) suggested a near-complete sampling of the bacterial diversity (Fig. S1). Relatively low Shannon diversity indexes of tailings (from 3.55 to 4.71), which were much lower than that reported in antimony-rich tailings sites (from 5.55 to 8.31) suggesting that the bacterial diversity was relatively low in these active tailings sites (Xiao et al., 2016).

Thiobacillus and the *Sulfurifustis* genera were shared between neutral tailings, which abundance was significantly different to that observed in acidic tailings sites ($p < 0.01$; Fig. S2 and Fig. S3) exhibiting significant pH differences ($p = 0.01$; Table S1). It has been reported that the iron- and sulfur- oxidizing bacterium *Thiobacillus*, which is known as acid tolerant, is capable of pyrite transformation and mineral sulfide moiety utilization (Norris et al., 2000). In addition, the enzymatic reactions and redox proteins involved in the oxidation mechanisms of *Thiobacillus* genus have been characterized. They include sulfur dioxygenase, sulfite-Cytc oxidoreductase, Fe(H)-Cytc oxidoreductase, and cytochromes c-552(s) (Yamanaka, 1996; Lim et al., 2015). The activity of *Thiobacillus* may accelerate the acidification of tailings and oxidation of tailings metal(loid)s. Interestingly, there are only four studies about *Sulfurifustis*, a sulfur oxidizer that grows optimally pH 6.8 to 8.2 (Kojima et al., 2015; Kojima et al., 2016; Umezawa et al., 2016; Issotta et al., 2018). It has been determined that *Sulfurifustis* genus harbor genes involved in sulfur oxidation (such as *aprAB* and *hdrAACB*) and inorganic carbon fixation pathways (such as *rbcL* and *rbcS*) (Umezawa et al., 2016), suggesting that the conditions prevailing in the tailings were favorable for sulfur and carbon metabolism.

We demonstrated that the distribution of microorganisms in tailings sites was influenced by high concentration of potentially toxic contaminants such as metal(loid)s and flotation reagents (Liu et al., 2019). The most fundamental mechanisms for generating and maintaining

ecological diversity in the tailings are driven by the environmental factors (Zhou and Ning, 2017). Spearman correlation analysis revealed that the abundances of bacterial genera were distributed within three main clusters, which were significantly correlated with environmental parameters ($p < 0.001$; Fig. 2). The first group showing significant positive correlation with TOC and Mn concentrations, was mainly composed by *Sphingomonas* ($p < 0.001$; Fig. 2). *Sphingomonas* possesses the genes involved in the biphenyl, naphthalene, phenanthrene, and m-xylene degradation, and metal-resistance (Kim and Zylstra, 1999; Reith et al., 2012), suggesting it probably plays a major role in the transformation of metal(loid)s and degradation of organic flotation reagents in the active tailings sites. The second group, showing significant positive correlation with Zn, Pb, TP, and Cd, included *Sulfurifustis*, *Iamia* and four unclassified genera ($p < 0.001$; Fig. 2). *Sulfurifustis* was dominant in neutral tailings sites (8.6%; Fig. 1). It is involved in sulfur oxidation and inorganic carbon fixation (Umezawa et al., 2016). The *Iamia* genus showed low abundance at the tailings sampling sites ($< 0.1\%$; Fig. 1). Its ecophysiology is poorly understood because only a few species have been isolated and identified. Additional functional information is required to understand its role in mine tailings (Kurahashi et al., 2009). The third group, showing significant positive correlations with Cu, included *Gaiella*, *Bryobacter*, and four unclassified genera (Fig. 2). *Gaiella* was dominant in the neutral tailings sites (8.6%; Fig. 1). Its role in tailings is poorly understood with only a handful of cultured species identified (Albuquerque et al., 2011). Again, further information is required to understand its role in mine tailings evolution. *Bryobacter* genus is known to contribute to V^{5+} reduction as a function of dissimilatory Fe^{3+} reduction (Hao et al., 2018). Thus, it is likely that these genera play an important role in the early stages of metal transformation and acidification of the nonferrous tailings.

Negative correlations were identified between bacterial population abundances and

environmental parameters: *Altererythrobacter* with Zn, Pb, TP, and Cd concentration as well as *Lysobacter* and *Klebsiella* with As concentration ($p < 0.001$; Fig. 2). The whole genome sequence analysis for *Altererythrobacter* showed that it exhibited strong resistance to several organic solvents (such as acetonitrile and DMSO) and metals (such as Cu, Cd, Mn, and Zn) (Hong et al., 2019). Its presence in the tailings may suggest that *Altererythrobacter* plays an important role in the metal(loid) enrichment of tailings. *Lysobacter* harbors eight genes corresponding to arsenic resistance in its genome, including the arsenic resistance (*arsH*) and the arsenical resistance regulator (Liu et al., 2015). It has been shown that As release is enhanced under more acidic pH coupled with dissolved O_2 and Fe^{3+} , further increased by microbial activities (Kim et al., 2002; Paikaray, 2014). Since *Klebsiella* genera harbors nitrite reductase (*nir*), nitric oxide reductase (*nor*), assimilatory nitrate reductase (*nas*) and respiratory nitrate reductase (*nar*) genes, it is likely that it is involved in nitrogen cycle in tailings playing indirectly an important role in microorganism adaptation to the physical-chemical parameters prevailing in active nonferrous metal(loid) tailings (Pal et al., 2015).

< insert Fig. 2 >

3.2 Metagenomic insights into the selection of multi-metal resistance genes

Microorganism possessing various resistance genes to deal with the extreme conditions could be changed with different environment factors (Yang et al., 2014). In comparison with uncontaminated environments, the acidic and neutral tailings sites studied represented extreme habitats (Table S1). To illustrate and validate the most representative tailings sites for metagenomic analysis of microbial communities, microPITA analysis was used based on the 16S rRNA sequencing data (Fig. S4). Among these samples, T1 and T4 exhibited high ecological dissimilarity and were selected for metagenomic analysis (Fig. S4). To reveal the

genes and metabolic capabilities involved in metal tolerance at the active nonferrous metal(loid) tailings, a metagenomic analysis for T1 and T4 was performed. The metagenome analysis revealed that *Thiobacillus* and *Gaiella* were dominant in the microbial community of the T1 and T4 tailings sites (Fig. S2), which was in accordance with the 16S rRNA gene sequencing analysis (Fig. 2). As well, *Thiobacillus* showed significant differences between the acid and neutral tailings sites ($p < 0.05$; Fig. S3). The microorganisms inhabiting such adverse ecosystems possess metabolic capabilities allowing to resist and tolerate metal(loid)s toxicity (Duran and Tamás, 2010; Liu et al., 2019). Approximately 94% of the 6968 KEGG Orthologs (KO) was assigned (Table S3). In addition, the KOs belonging to the biosynthesis of amino acids, carbon metabolism, purine metabolism, ABC transporters, and two-component systems were the top five enriched functional attributes in both the T1 and T4 metagenomes (Table S3), and showed significant differences ($p < 0.001$; Fig. S5). The microorganism metabolic activities are the causes for the dissolution of minerals, and can also contribute to the biosorption, bioprecipitation, extracellular sequestration, transport, and/or chelation of tailings metal(loid)s (Haferburg and Kothe, 2010).

To explore the microbial metabolic capacities to tolerate the continuous accumulation of metal(loid)s in tailings, the multi-metal resistance genes (MRGs) were examined (Fig. 3). From the metagenomic data, 43 KOs involved in MRGs were identified, including As, Cu, Mn, and Zn metabolism pathways (Fig. 3 and Table S4). The number of MRG subtypes was lower than that previously reported for lakes (218 and 166 MRG subtypes, respectively) (Yang et al., 2019), which has been contaminated by nonferrous metal(loid) tailings. The tailings KOs were mainly related to transport, transcriptional, and resistance-associated pathways (Table S4 and Table S5). It is likely that the microorganisms exposed to toxic compounds (metal(loid)s and FRs) in active tailings represent a short-term exposure leading to a lower MRGs accumulation

than that reported for long-term exposure (Sven et al., 2015). Genes encoding As, Cu, Mn, and Zn cell transportation and the related regulators were detected in tailings metagenomes (Fig. 3). The *arsAB*, *arsC*, *aoxAB*, and *arsH* genes, known to be involved in arsenic transformation processes-featuring reduction, oxidation and resistance mechanisms (Xiang et al., 2008; Escalante et al., 2009), were detected in the tailings metagenomes (Fig. 3). The detection of such As-resistance related genes suggested that the indigenous microorganisms possessed the capacity to influence the As speciation in the tailings. The presence of these As-resistance genes together with the genes involved in copper (*copABC* and *cusAF* genes) and zinc resistances (*znuABC* genes) in the tailings metagenomes (Fig. 3; Carlos and Felix, 1994), suggested that microorganisms possessing genetic determinants for metal(loid) resistance were selected by the high metal(loid) concentrations prevailing in the tailings.

< insert Fig. 3 >

Interestingly, genes encoding the arsenite-transporting ATPase (*arsA*) and the copper resistance (*copC*) were significantly different between T1 and T4 tailings ($p < 0.05$; Table S5). The analyses of the taxonomic and functional relationships showed that *Thiobacillus* and *Nocardioides* genera were the most significant contributors to the different MRGs (Fig. 4a). These two genera harbored the As/Cu resistance gene (Table S4). It has been demonstrated that *Thiobacillus* could encode as many as 17 possible metal resistance systems, such as Ni^{2+} (*nreB*), Pb^{2+} (*pbrT*), and Cr^{6+} (*chrA*) (Beller et al., 2006), and can acquire metal resistance genes by a combination of gene gains and losses, horizontal gene transfer, and gene duplication (Gillan et al., 2015). *Nocardioides* exhibits arsenic transformation ability, and showed oxidation of As^{3+} prevailing over As^{5+} reduction (Paikaray, 2014; Bagade et al., 2016). At the same time, *Nocardioides* carries Cu efflux-mediated metal resistance genes (Kaplan et al., 2019). Results suggested that *Thiobacillus* and *Nocardioides* might play an important role for the adaptation

of microorganisms in the active nonferrous tailings.

< insert Fig. 4 >

3.3 Co-selection of antibiotic resistance genes and MRGs

In the present study, across the dataset, the most abundant ARGs in both tailings were the bacitracin resistance genes (Fig. 5). They were found to be significantly different at the acid tailings (T1) in comparison to the neutral tailings (T4) ($p < 0.05$; Fig. S6 and Table S6). The second most ARGs class was the efflux pump system (19.1%-20.5%; Fig. 5), which have been demonstrated to be responsible for the extrusion of the antibiotic outside the cell (Marquez, 2005). However, resistance genes obtained by mutation and knockout process were accounted for 22.8% at active tailings sites (Fig. 5). The mutation in ARGs might play a key role for surviving antibiotic-mediated killing and control of antibiotic-resistant pathogens (Lin et al., 2010). The main genera carrying ARGs were found to be low abundant ($< 0.1\%$), and mostly affiliated with low abundant unclassified bacteria and “others” rather than known taxa (Fig. 4b). The “unidentified populations” represented over 25% of total populations (Fig. 5, Fig. S6 and Table S6), which is consistent with previous studies indicating that the contamination of metal(loid) and flotation reagents in tailings sites had vast unidentified populations and microbial resources (Liu et al, 2019; Xiao et al., 2016). Coupled with the fact that nonferrous tailings were contaminated by high concentration of metal(loid), the lower abundance of ARGs in comparison to that of MRGs was expected since tailings represent a different environment than that where antibiotics are frequently used, such as pig-breeding (Zhao et al., 2019). But the observation of ARGs in tailings suggested that tailings also represent a ‘pollutant source’ of ARGs. As well, the *abca* resistance type was the most abundant ARGs (Fig. 5). Such observation suggested that the low abundant genera represent a ‘seed bank’ of ARGs for

microorganisms able to survive to the high concentration of metal(loid)s and flotation reagents observed in the tailings.

< insert Fig. 5 >

The co-occurrence pattern between ARGs and MRGs was explored by network analysis based on Spearman's correlation ($r > 0.7$, $p < 0.05$; Fig. 6). Co-occurrence analysis for the microorganism KOs in tailings showed that integron gene cassettes of MRGs (copper export genes, *copA*) and ARGs (such as *baca*, *mdete* and *mdtp*) were correlated (Fig. 6, Fig. S7 and Table S8). It has been demonstrated that *baca*, *mdete* and *mdtp* resistance genes carried with integron gene cassettes in the same mobile genetic elements could be responsible for the co-selection of ARGs because of metal(loid) contamination (Lin et al., 2010; Yang et al., 2019). As well, genes related to Cu/Zn exporting and transport, such as *zntA* were correlated with ARGs (*mdte* and *mdtp*; Fig. 6 and Table S8). The accumulated toxic metal(loid)s may in some cases provide stronger and more persistent selective pressures for antibiotic resistance as compared to antibiotic residues from agricultural soils (Zhao et al., 2019). Although our study was not specifically designed to compare the relative importance of different co-selection mechanisms, the co-selection of MRGs and ARGs was clearly demonstrated (Fig. 6), which was consistent with previous study indicating that zinc-resistance and carbapenem resistance were acquired by co-regulation (Perron et al., 2004). Our study suggested that tailings metal(loid) contamination co-selected ARGs that may be explained by horizontal gene transfer between microorganisms.

< insert Fig. 6 >

3.4 Microbial metabolic potential to the extreme tailings

In order to determine organic metabolism, Fe-S metabolism, and acid stress functional

metabolic pathways allowing to cope with the extreme tailings conditions, KOs based on the Metagenomic database were examined (Fig. 3). Previous research on residual flotation reagents (FRs) in active nonferrous metal(loid) tailings showed that heterocyclic compounds, benzodiazepines, and polycyclic aromatic hydrocarbons (PAHs) were the main intermediate metabolites of FRs (Table S7), which were more toxic than their parents (Liu et al., 2018; Zhu et al., 2018). Functional metabolic genes for benzenes (such as *benABC*, *bcrABCD*, *boxAB*) were detected (Fig. 3). Genes for the degradation of toluene (*bbsEF*) were only enriched in the neutral nonferrous metal(loid) tailings (Fig. 3). As well, multiple genes belonging to the benzoyl-CoA 2,3-epoxidase subunit B, including the *Rta*, *Mrm*, *Ccro*, and *BBta* genes, were neutral-enriched, and involved in *Labilithrix*, *Cupriavidus*, and *Tardiphaga* genus (Table 1). Results suggested that the microorganisms have developed functional pathways to degrade the FRs, which in turn contribute to the production of non-biodegradable intermediate organics in active tailings (Zhu et al., 2018).

< insert Table 1 >

Considering to the fact that mining and processing waste at active tailings sites had the potential to generate acid mine drainage (Volant et al., 2012), genes involved in iron-sulfur metabolism were analyzed (Fig. 3). Genes related to iron-sulfur metabolisms were mainly including sulfur oxidase, adenylylsulfate reductase, dissimilatory sulfite reductase (*dsr*), and an Fe-S cluster assembly genes (Table S3). Sulfate reduction genes (*dsr*) that codes enzymes catalyzing the reduction of sulfite to sulfide and required by all sulfate reducers (Geets et al., 2006), showed relatively high abundance (genes number > 18, Table 1). The *cyt*-like genes could participate in iron oxidation, which are common in AMD environments (Tan et al., 2019). Moreover, identification of *iro*-like genes in tailings sites may demonstrate that microbial communities of tailings have the potential of iron oxidization (Fig. 3). *iro*-like genes could

accelerate the acidification of tailings, especially under acidic conditions ($\text{pH} < 4.5$) (Chen et al., 2013). *Conexibacter* and *Solirubrobacter* were the main contributors to distribution of Fe-S metabolism at tailings sites (Table 1) and contributed significantly to the functional pathways of neutral active tailings ($p < 0.05$; Fig. 4). But there was little functional gene information for the *Conexibacter* and *Solirubrobacter* genera, which were always isolated from the soil environment (Kim et al., 2007; Wiker et al., 2019). The different types of iron-sulfur cluster assembly genes may provide a metabolic versatility for the survival of microbial communities in active tailings sites, where iron-sulfur biooxidation might strengthen their ability to produce AMDs (Chen et al., 2013; Tan et al., 2019).

Additionally, genes encoding the acid stress chaperone (*HdeAB*) were detected, and specific at the acid tailings ($\text{pH} = 2.0$; Fig. 3 and Table S3). Previous studies have demonstrated that *HdeAB* prevent the acid-induced aggregation of bacterial periplasmic extracts and of several model substrate proteins, such as alcohol dehydrogenase (ADH) (Kern et al., 2007). And *HdeB* is more efficient than *HdeA* in preventing periplasmic-protein aggregation at pH 3 (Kern et al., 2007). In order to adapt to extremely acidic habitats, microorganisms have been known to harbor enzymes and/or chemicals capable of binding and sequestering protons (Chen et al., 2013). Functional abundance profile analysis based on the metagenomic data collected for this study attributed the acid stress genus to ‘unclassified’ (Table 1). Such observations made at the functional pathways of microbial community level may predict the microbial response to tailings disturbance. However, metabolic pathways in the present study uncovered the whole organic/Fe-S metabolic and acid stress pathways (Fig. 3), suggesting that the metabolic pathways were incomplete. More complete metagenomic analysis is needed in the future to help for further understanding Fe-S metabolism and acid stress in the active tailings.

4 Conclusions

Our study provides a new insight into the bacterial functional metabolisms in active nonferrous metal(loid) tailings that receive uninterrupted discharges of mine waste. It is likely that the sulfur-oxidation of *Thiobacillus* and *Sulfurifustis* genera accelerate the acidification of tailings. Metagenomic analysis revealed the presence of MRGs such as arsenite and copper resistance genes. *Thiobacillus* and *Nocardioides* genera were the main contributors carrying the MRGs, which were differentially distributed among the tailings sites. Moreover, we observed the co-selection of MRGs and ARGs (such as bacitracin and zinc resistances genes) even in the absence of antibiotic selection pressure. Our findings caution that the active tailings could be a dissemination source of tailings resistomes. As well, the short-term exposure of microorganisms to high toxic metal(loid) active tailings contribute to the aggravation of tailings acidification and production of non-biodegradable intermediate organics. Our study suggests that careful monitoring of active tailings sites before they are abandoned should be considered as early as possible, to enable the timely management of any potential environmental risks present at these sites.

Acknowledgments

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Figure legends

Fig. 1. Map of the sampling sites (left panel) and their bacterial community composition (right panel). The four sites (T1 to T4, represented by colored circles) are located close to Hechi City and Nandan County, Guangxi, China. The relative abundances of the major bacterial genus are presented for each tailings using 16S rRNA MiSeq sequencing data-set. Sequences assigned into the unclassified or no_rank genera are not shown.

Fig. 2. Spearman correlation between geochemical parameters and the 50 most abundant bacterial genera (16S rRNA MiSeq sequencing data-set were used). Red, positive correlations. Blue, negative correlations. White, non-significant correlations. The bar at the right bottom was represented the correlation coefficient. * $0.01 < p \leq 0.05$; ** $0.001 < p \leq 0.01$; *** $p \leq 0.001$.

Fig. 3 Potential microbial metabolic pathways to cope with the tailings stress detected in the tailings metagenomes. The metabolic pathways include carbon/nitrogen/sulfur metabolism, metal(loid)s resistance, and acid stress/organic tolerance. Genes marked with blue square were specifically found in the neutral tailings (T4). Detailed information on the genes is provided in Supplementary Table S3.

Fig. 4. Barplots of species contribution analysis to the microbial metabolism (a) and antibiotic resistance (b). Red, positive correlations. Blue, negative correlations. White, non-significant correlations.

Fig. 5 Venn diagram (center) showing the overlap degree of relative abundances of antibiotic resistance genes based on the Metagenomic data. a, AR (Antibiotic Resistance) genes directly resist the antibiotic; b, AT (Antibiotic Target) genes were target-binding; c, ABS (Antibiotic Biosynthesis) gene represent antibiotic biosynthesis related genes; d, AS (Antibiotic Sensitive) gene obtained by the process of mutation, knockout, etc.

Fig. 6 Co-occurrence network analysis of antibiotic resistance genes (ARGs, a) and KEGGs (b) between tailings sites based on Metagenomic data. The spearman correlation coefficient > 0.7 and p -value < 0.05 were showed in present figure. The most abundant ARGs and KEGGs (50 most abundant) were used for analysis. Each node indicates an ARG and KEGG. The green nodes without names (a) represent the ARGs functions related to *abca* resistance type. The black code (a), genes of *mdte* and *mdtp* were involved in resistance-nodulation-cell division transporter system and multidrug resistance efflux pump. The pink code (b) represented genes related to metal(loid)s, such as *copA* gene, which encode Cu^+ exporting ATPase. More information on KEGGs numbers is provided in Supplementary Table S8.

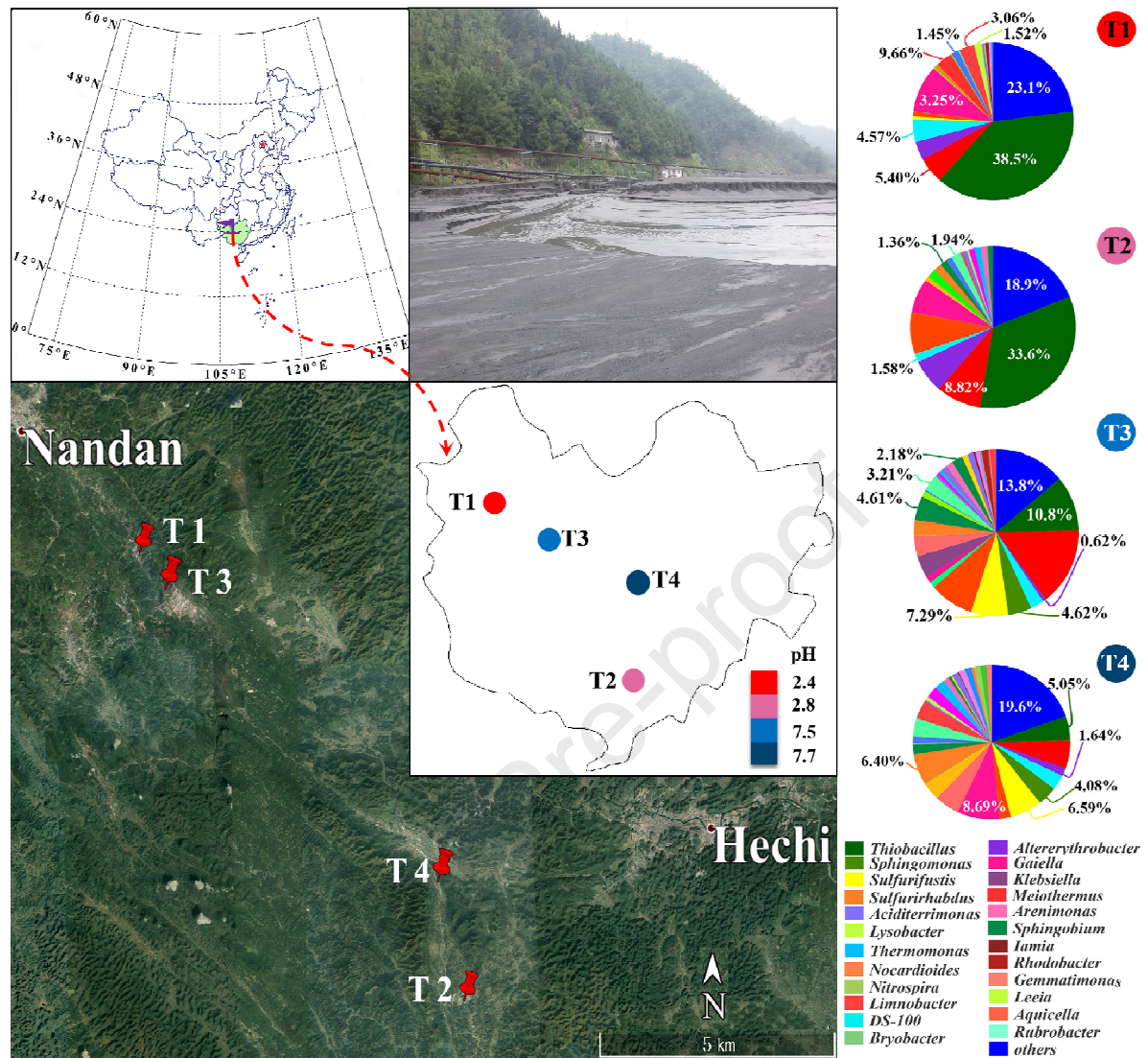


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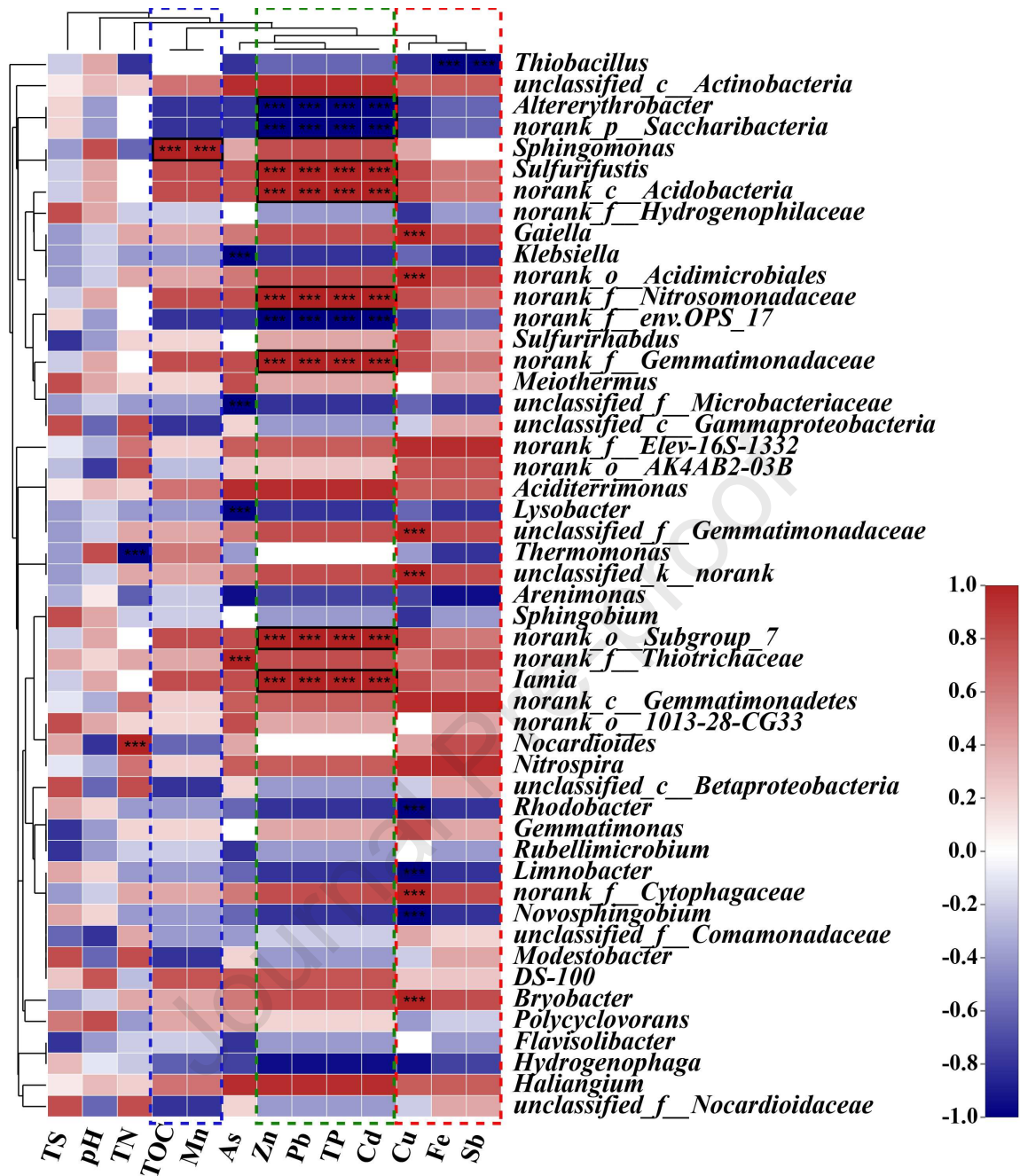


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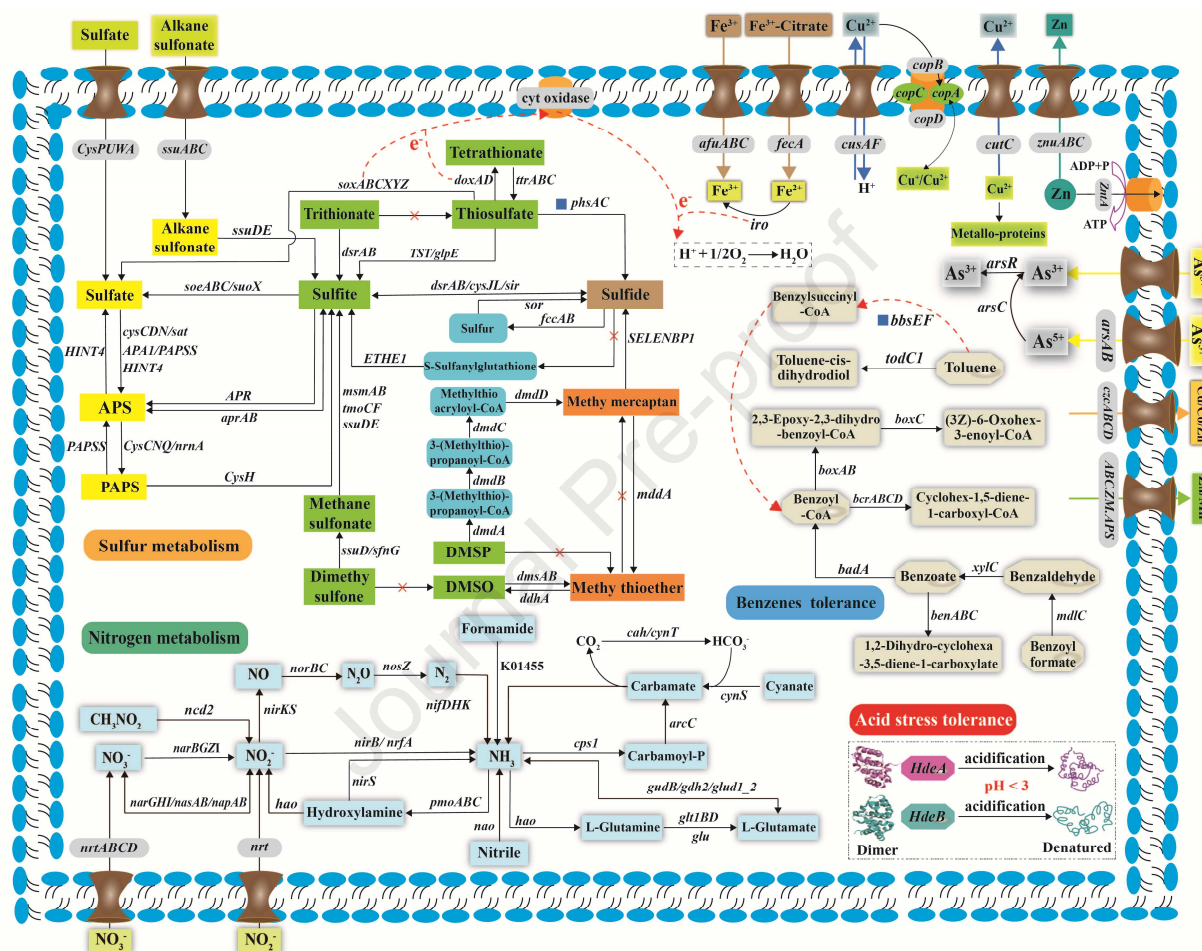


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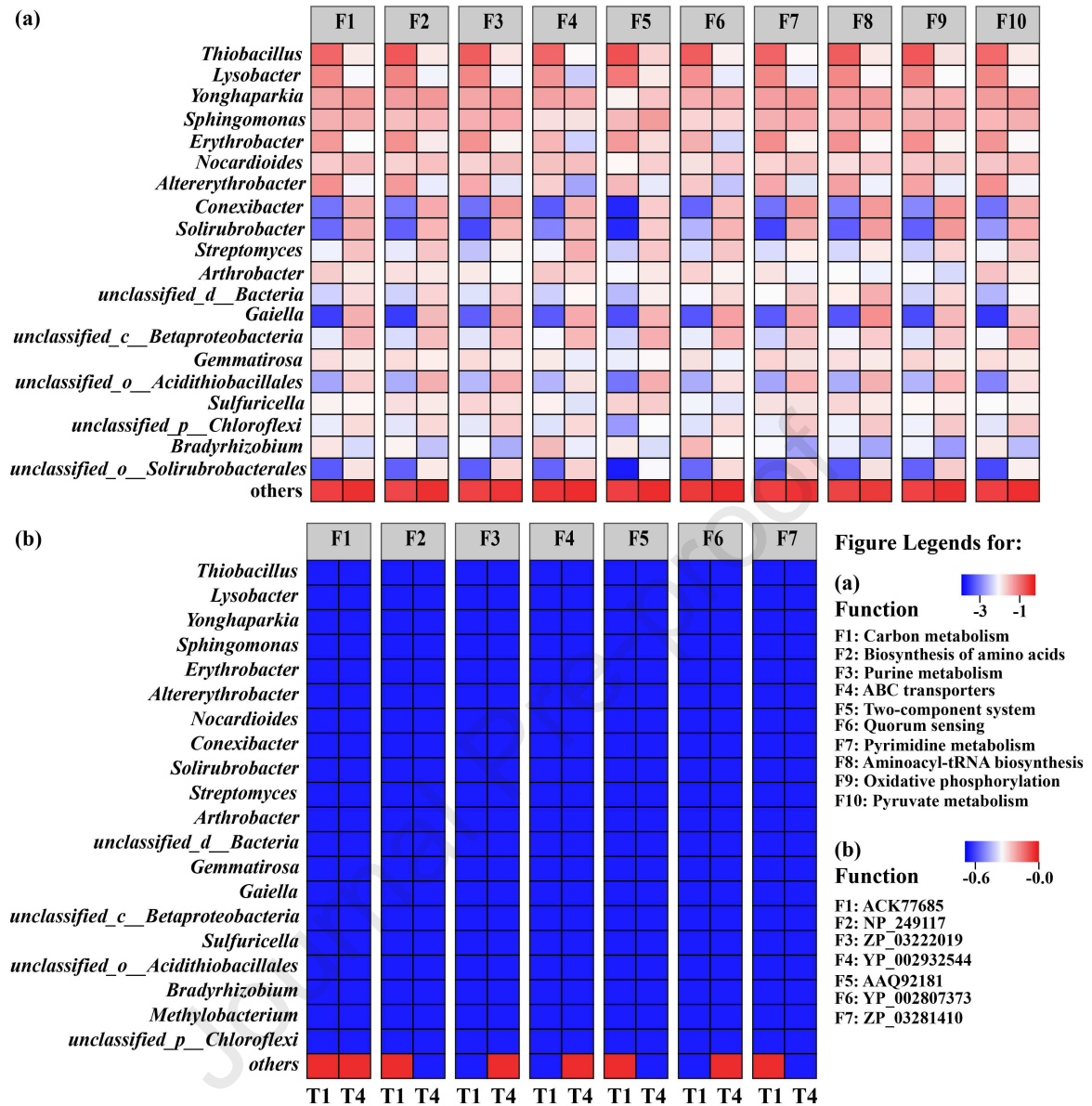


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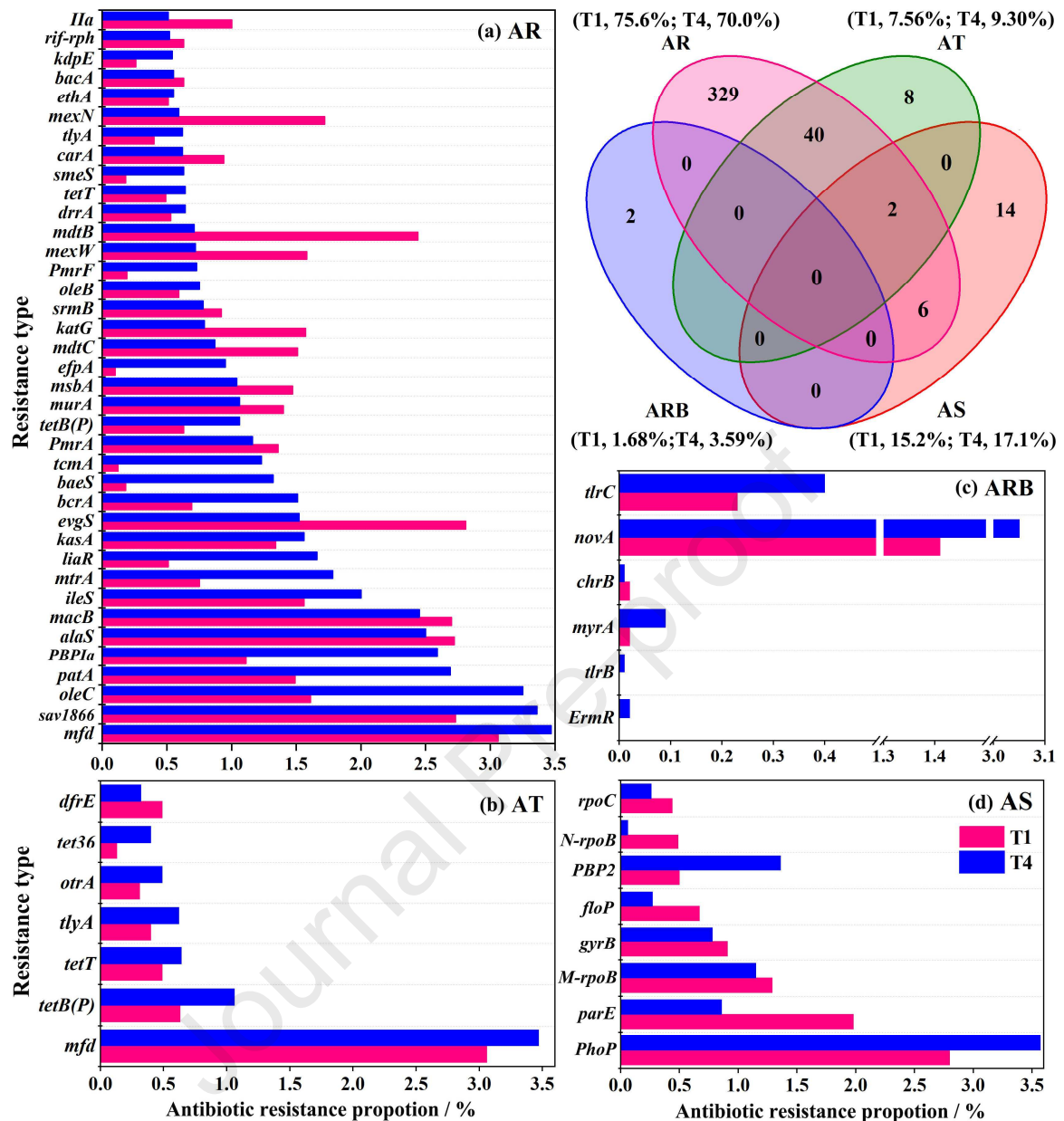


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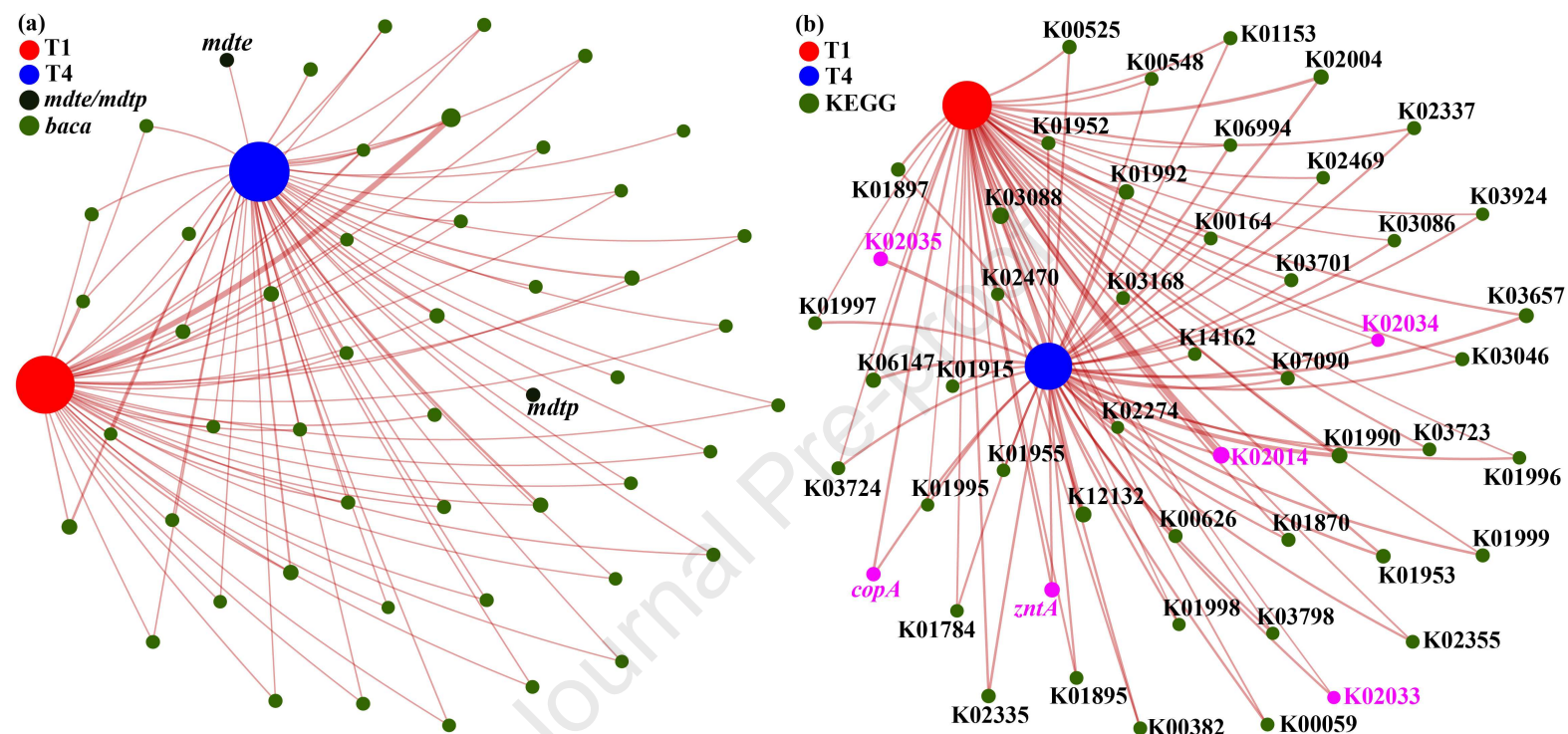


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Table 1 Summary of genes/enzymes involved in microbial carbon metabolism, Fe-S oxidation, and acid stress in T1 and T4 metagenomes.

Key metabolic function	T1		T4	
	No. gene	Main associated genome bin	No. gene	Main associated genome bin
Autotrophic carbon fixation				
TCA cycle	33	<i>Thiobacillus, Lysobacter</i>	51	<i>Thiobacillus, Gemmatirosa</i>
acetyl-CoA	49	<i>Thiobacillus, Lysobacter, Sphingomonas</i>	82	<i>Thiobacillus, Tysobacter, Yonghaparkia, Gaiella</i>
Xenobiotics biodegradation				
Aminobenzoate degradation	0		2	<i>Longispora, Micromonospora</i>
Benzoate degradation	11	<i>Herminiimonas, Sphingomonas,</i>	16	<i>Labilithrix, Cupriavidus, Tardiphaga, Mycobacterium</i>
Fatty acid degradation	17	<i>Streptomyces, Rhodopseudomonas, Methylobacterium</i>	8	<i>Gaiella, Erythrobacter, Actinomadura</i>
Fe-S metabolism				
sulfite oxidase	0		1	un_classified
sulfur-oxidizing	4	<i>Thiobacillus, Polaromonas</i>	2	<i>Sulfuritalea</i>
adenylylsulfate reductase	1	<i>Sulfuritalea</i>	1	<i>Thiodictyon</i>
dissimilatory sulfite reductase	18	<i>Altererythrobacter, Lysobacter, Thiobacillus, Sulfuritalea</i>	23	<i>Arthrobacter, Thiobacillus</i>
Fe-S cluster assembly protein	2	<i>Burkholderia, Nocardioidea</i>	6	<i>Gemmatirosa, Conexibacter, Solirubrobacter</i>
thiol:disulfide interchange	3	<i>Bradyrhizobium</i>	1	<i>Sulfuricella</i>
Acid stress	2	un_classified	1	un_classified

Highlights

- Microorganism from four typical non-ferrous active tailings were investigated
- MRGs and ARGs were co-selected by high concentrations of metal(loid)s tailings
- *Thiobacillus* and *Nocardioides* were the main contributors of MRGs distribution
- The low abundant genera (< 0.1%) were the main contributors for ARGs distribution
- Short-term metal exposure contributed to the aggravation of tailings acidification

Author Statement

Dear Editor,

On behalf of my co-authors, I declare that the described work is original research that has not been published previously, and is not under consideration for publication elsewhere (in whole or in part). No conflict of interest exists in the submission of this manuscript. All the authors listed have approved the attached manuscript.

We also made the modification according to the comments of Editors and Reviewers. We appreciate your acceptance of our manuscript. If you have any queries, please contact me. Thank you and with best regards.

Yours sincerely,

Jun Yao

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: